

The empirical recurrence for OEIS A240787: recovering a conjecture that is not written down

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Abstract

OEIS A240787 records “Empirical recurrence of order 94”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 788$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 94, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 94 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A240787 is “Number of $6Xn$ 0..1 arrays with no element equal to fewer vertical neighbors than horizontal neighbors, with new values 0..1 introduced in row major order.”. It has offset 1 and begins

32, 361, 2187, 15332, 106411, 746346, 5228820, 36701371, ...

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 11:07:44, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 788$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 788$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1576$ exact terms of the model returns order 94 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	20	c_{25}	3297853972	c_{49}	−1507567565841	c_{73}	−32089643154
c_2	−166	c_{26}	−5100687096	c_{50}	−5355520055839	c_{74}	25677996187
c_3	764	c_{27}	9588316681	c_{51}	−7436947953895	c_{75}	29201158525
c_4	−2035	c_{28}	−19043678545	c_{52}	−5928759949837	c_{76}	16812898333
c_5	2341	c_{29}	24235577659	c_{53}	−1636241814537	c_{77}	6686853108
c_6	3581	c_{30}	−12640068370	c_{54}	4216359267627	c_{78}	1236311162
c_7	−20276	c_{31}	−11986778399	c_{55}	7659737426394	c_{79}	−528781687
c_8	40792	c_{32}	30942692531	c_{56}	9196199764950	c_{80}	−590043236
c_9	−42877	c_{33}	−10562870016	c_{57}	6923080103233	c_{81}	−304322434
c_{10}	−11037	c_{34}	16086451747	c_{58}	2997291052154	c_{82}	−92870523
c_{11}	221937	c_{35}	−54603381686	c_{59}	−1828998483695	c_{83}	−9474601
c_{12}	−466951	c_{36}	132875870969	c_{60}	−5606123392522	c_{84}	6370831
c_{13}	−1283077	c_{37}	−181101828091	c_{61}	−6865574701417	c_{85}	3833805
c_{14}	10621040	c_{38}	81744989674	c_{62}	−5633764431616	c_{86}	1781503
c_{15}	−33612877	c_{39}	−200157350857	c_{63}	−2471717430023	c_{87}	351946
c_{16}	70510514	c_{40}	19230847593	c_{64}	690291601188	c_{88}	−16682
c_{17}	−120987854	c_{41}	46698956176	c_{65}	2635108015062	c_{89}	−17124
c_{18}	174776066	c_{42}	−366465773992	c_{66}	2838090480442	c_{90}	−6182
c_{19}	−147589041	c_{43}	−210833046582	c_{67}	1824963162476	c_{91}	580
c_{20}	−69333103	c_{44}	27721993661	c_{68}	558000876662	c_{92}	−472
c_{21}	414841712	c_{45}	1422837186034	c_{69}	−302915797420	c_{93}	−310
c_{22}	−654963671	c_{46}	1758718773725	c_{70}	−552583726774	c_{94}	28
c_{23}	935992873	c_{47}	2753559558998	c_{71}	−417114199571		
c_{24}	−1898336511	c_{48}	892204278475	c_{72}	−187232873228		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 94, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $94 + 4$ terms removed returns order 94 as well, so $\deg q_{\min} = 94 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $94 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 94 that this sequence satisfies — and that family turns out to have exactly one member.

References

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